# PROJECT OVERVIEW AND METHODOLOGY GUIDE

## DNS-Based Covert Channel Detection Using Machine Learning

*A Comparative and Cross-Dataset Analysis of DNS Exfiltration and DNS-over-HTTPS Traffic*

*Figure 1. High-level architecture of the research framework.*

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*Purpose: shared understanding, implementation walkthrough, experiment interpretation, and viva preparation.*

## How to Use This Document

This document explains the research problem, intended methodology, current implementation context, experiments E1–E8, feature analysis, metrics, visualizations, limitations, and the evidence that can responsibly be presented. It is written for teammates who need to understand both what the system does and what its results do—and do not—show.

**Evidence note** This guide is based on the research-paper draft "DNS-Based Covert Channel Detection Using Machine Learning: A Comparative and Cross-Dataset Analysis of DNS Exfiltration and DNS-over-HTTPS Traffic" and the experiment summaries, dashboard outputs, and audit notes shared in this conversation. The local project repository was not directly inspected during creation of this document. Therefore, implementation details and metrics described as reported should be checked against the current run-specific files before formal submission.

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# 1. Executive Summary

The project studies machine-learning-based detection of DNS-related covert communication and data exfiltration. Conventional DNS (Do53) can expose the queried domain and DNS fields to a network observer. DNS-over-HTTPS (DoH) carries DNS messages inside encrypted HTTPS connections, hiding query content from an ordinary passive observer and shifting detection toward observable flow, timing, size, and TLS-related characteristics.

The research framework compares six classifiers—Logistic Regression (LR), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest (RF), XGBoost (XGB), and Multi-Layer Perceptron (MLP). Eight experiments organize the work: two within-dataset baselines, two cross-protocol transfer directions, a feature-group ablation, top-K feature selection, external benchmark evaluation, and DNSTunnel2026 validation/transfer analysis.

The project combines metric reporting with feature analysis, SHAP explanations, saved run configurations, held-out predictions, plots, and a dashboard. The important scientific challenge is generalization: high performance on a random split from one dataset does not prove that the model will work on unseen domains, sessions, campaigns, capture periods, or different protocols.

**Main takeaway** The strongest defensible project story is not “every model is perfect.” It is that a comparative framework is being used to study within-dataset detection, cross-dataset/protocol transfer, feature relevance, and the limits of random-split evaluation.

## Project at a glance

| Item | Description |
|---|---|
| Problem | Detect traffic labelled as DNS/DoH exfiltration while controlling false alarms. |
| Primary datasets | UFU-Do53-EXF and UFU-DoH-EXF. |
| External datasets | CIC-Bell-DNS-EXF-2021, CIRA-CIC-DoHBrw-2020, DNSTunnel2026. |

| Item | Description |
|---|---|
| Models | LR, KNN, SVM, RF, XGBoost, MLP. |
| Experiments | E1–E8: baselines, transfer diagnostics, feature studies, benchmark/external validation. |
| Analysis | Precision, recall, F1, ROC-AUC, PR-AUC, FPR, confusion matrices, feature importance, SHAP. |
| Main risk | Dataset/campaign-specific patterns can inflate random-split performance. |
| Current status caveat | Reported results vary by run and artifact. Use exact run IDs and verify files before submission. |

# 2. Problem Statement and Background

## 2.1 Problem statement

Organizations depend on DNS, so blocking all DNS communication is usually impractical. An attacker may abuse DNS queries to move encoded fragments of unauthorized data to infrastructure under their control. DoH creates an additional monitoring challenge because DNS messages are encrypted inside HTTPS. A detector that relies on readable domain strings may not have access to those strings when monitoring encrypted traffic.

The problem addressed by this project is to evaluate how effectively machine-learning classifiers can identify benign versus exfiltration-labelled traffic in conventional DNS and DoH datasets, and to investigate how detection performance and feature relevance change when the dataset, protocol, or feature representation changes.

## 2.2 Key definitions

| Term | Meaning in this project |
|---|---|
| DNS | The Domain Name System maps domain names to DNS records and related information. |
| Do53 | Conventional DNS, commonly associated with DNS over port 53; query names and DNS fields may be observable depending on capture position. |
| DoH | DNS over HTTPS. DNS messages are transported over HTTPS, which encrypts the content in transit. |
| DNS tunneling | Using DNS exchanges as a carrier for data or communications outside ordinary DNS use. |
| Data exfiltration | Unauthorized transfer of information out of a system or network. |
| Feature | A measurable input value used by a model, such as query length, TTL variability, packet size, or flow duration. |
| Label | The target class, for example benign or exfiltration, assigned by the dataset or label-generation procedure. |
| Generalization | The ability to perform on genuinely unseen examples, conditions, datasets, campaigns, or protocols. |

### 2.3 Why conventional DNS and DoH differ

*Figure 2. Observable feature spaces differ between conventional DNS and DoH.*

For Do53, lexical signals such as query length, character entropy, label composition, DNS record types, and TTL-related statistics may be available. For DoH, a passive observer generally cannot inspect the encrypted DNS query name, so detection often relies on traffic-level, flow-level, temporal, statistical, and observable TLS characteristics. The exact set depends on the dataset and where traffic was captured.

**Important distinction** DNS tunneling is a technique; data exfiltration is an objective. Malicious DoH traffic, DGA-related traffic, and DNS exfiltration are overlapping but not identical labels. Interpret each benchmark according to its own documented target label.

## 3. Research Gap, Objectives, Questions, and Hypotheses

### 3.1 Research gap

The research paper argues that many detection studies report strong results on an individual dataset, often under a random train/test split. Such scores do not automatically demonstrate robustness across a different resolver, network, collection period, attack tool, campaign, or protocol. This project is designed to go beyond a single within-dataset score by comparing models, studying feature groups, testing transfer in both directions, and inspecting model explanations.

### 3.2 Objectives

- Build a unified comparative framework for conventional DNS and DoH exfiltration-labelled datasets.
- Compare six established machine-learning classifiers under documented evaluation conditions.
- Measure precision, recall, F1, ROC-AUC, PR-AUC, false-positive rate, and computational cost where available.
- Investigate whether models trained on one protocol or dataset can transfer to another.
- Assess the role of feature groups and compact top-K feature sets.
- Use feature importance and SHAP to explain what information trained models use.
- Use external benchmarks to provide additional evidence, while respecting their label and feature definitions.
- Create a reproducible dashboard/report workflow that traces results to saved artifacts.

### 3.3 Research questions from the paper

- RQ1: How effectively can different ML models detect conventional DNS-based data exfiltration?
- RQ2: How effectively can the same models detect DoH-based data exfiltration?
- RQ3: How well do models trained on conventional DNS exfiltration generalize to DoH exfiltration?
- RQ4: How well do models trained on DoH exfiltration generalize to conventional DNS exfiltration?
- RQ5: Which feature groups remain informative when DNS content becomes encrypted?
- RQ6: Which ML approach gives a useful balance among recall, precision, F1, PR-AUC, FPR, and computational cost?
- RQ7: Which features contribute most strongly to model decisions, and do they remain stable across datasets?

### 3.4 Research hypotheses

| Hypothesis | Statement | Evidence required |
|---|---|---|
| H1 | Models will achieve strong performance in within-dataset DNS and DoH detection. | Valid held-out E1/E2 evaluations, correct labels, no leakage, and suitable metrics for imbalanced data. |
| H2 | Cross-dataset/protocol performance will be lower than within-dataset performance because feature distributions and observable information differ. | Comparable within-data and valid cross-data results; proxy experiments alone are insufficient. |
| H3 | Flow-level and temporal features will be more transferable to DoH than plaintext DNS lexical features. | Controlled feature comparison using consistently defined measurements and valid target labels. |
| H4 | The most important features identified by explainable ML will differ between conventional DNS and DoH environments. | Comparable model explanations and stability checks across appropriate runs/datasets. |

**How to conclude** A hypothesis may be supported, partially supported, not supported, inconclusive, or not evaluated. Missing or invalid evidence is not evidence that a hypothesis is false.

## 4. Datasets and Feature Spaces

The research paper uses a layered data strategy: paired UFU datasets are primary because they directly represent conventional DNS and DoH exfiltration; established 2020/2021 datasets are external benchmarks; DNSTunnel2026 is an additional external-validation target when data, labels, and features allow a defensible evaluation.

| Dataset | Published / reported size | Role | Key notes |
|---|---|---|---|
| UFU-Do53-EXF | 5,351,064 records; 29 features; 4,812,119 benign and 538,945 exfiltration. | Primary Do53 | About 10.1% positive in the published count. Includes lexical, DNS and domain-context features. |
| UFU-DoH-EXF | 1,048,575 records; 28 features; 1,016,438 benign and 32,137 exfiltration. | Primary DoH | About 3.1% positive in the published count. Includes encrypted-flow/timing/statistical features. |
| CIC-Bell-DNS-EXF-2021 | A recent project note reports 18 Stateless and 18 Stateful CSVs; 1,000 deduplicated records per representation in one evaluation. | External DNS benchmark | Stateless: 12 numeric lexical features. Stateful: 20 numeric RR/TTL-related features. Folder-based labels need documented validation. |
| CIRA-CIC-DoHBrw-2020 | One project note reports 995 deduplicated L2 flows, split 796 train / 199 test. | External DoH benchmark | l2-benign.csv and l2-malicious.csv reportedly used. The task is benign DoH vs malicious DoH tunneling, not automatically exfiltration. |
| DNSTunnel2026 | The project notes describe train.csv and test.csv, with a reported 38-feature schema. | External validation | Counts, labels, feature definitions, and transfer compatibility must be confirmed from actual files. |

### 4.1 Published counts versus experiment counts

The large counts in the paper describe the published primary datasets. Experiment runs may use smaller samples—for example, the synchronized E1/E2 runs discussed later target a 15,000-row sample. A run must report the number of rows loaded, deduplicated, sampled, and retained after preprocessing. A result based on 15,000 sampled rows must not be described as training on every row of a 5.35-million-row dataset.

### 4.2 Where data may reside

The project notes state that dataset discovery may look outside icsproject, including a sibling directory such as `..\DNS_Covert_Channel_Project\data\raw\`. Therefore, an absent folder inside icsproject alone does not prove a dataset is unavailable. The configured discovery paths and actual loaded file paths should be checked and saved in the run manifest.

### 4.3 Why feature compatibility matters

The primary UFU CSVs have been reported to share zero exact non-label feature columns. This is not, by itself, a complete semantic proof that no equivalent measurements can be derived, but it prevents direct matrix transfer using the existing columns. Features must be comparable in definition, units, observation window and aggregation level—not merely have similar names.

**PCAP limitation** The shared E3/E4 diagnostic output reports incomplete DNS-360 PCAP archives. One output states Do53 is missing 2 of 34 parts and DoH is missing 16 of 66 parts; another checklist describes 11 missing DoH parts. This discrepancy must be reconciled from the actual archive inventory before the research paper states an exact count.

## 5. System Architecture and End-to-End Workflow

*Figure 3. Main data-to-results workflow.*

### 5.1 Components

| Component | Purpose | Expected output |
|---|---|---|
| Configuration / discovery | Resolves dataset paths, result directories, and metadata. | Traceable input paths and experiment configuration. |
| Dataset loader / audit | Reads records, checks columns, labels, counts, missing values and duplicates. | Validated dataset summary and usable records. |
| Preprocessing | Separates X and y, splits data, imputes missing values, encodes/scales as needed. | Train/test matrices processed without fitting transforms on test data. |
| Feature engineering / selection | Builds lexical, behavioral, flow, timing, or TLS-related features; ranks or selects features when required. | Model input features and selection metadata. |
| Model layer | Fits LR, KNN, SVM, RF, XGBoost or MLP. | Trained estimator and prediction scores. |
| Experiment controller | Implements E1–E8 designs and selects source/target datasets. | Run-specific outputs and status. |
| Metrics / explainability | Computes metrics, confusion matrices, feature importance, SHAP and plots. | JSON, prediction files, figures and interpretation summaries. |

| Component | Purpose | Expected output |
|---|---|---|
| Dashboard / report | Displays runs, datasets, hypotheses, outputs and downloadable reports. | Human-readable interface and research artifacts. |

### 5.2 Typical run artifacts

- Run configuration JSON: records the experiment, dataset, model and parameters used.
- Held-out predictions CSV: should preserve true labels, predicted classes and scores where available.
- Metrics JSON: stores evaluation metrics and associated run metadata.
- Plots: confusion matrices, ROC curves, precision-recall curves and comparisons when inputs are available.
- SHAP artifacts: global or local attribution data and plots for supported, fitted models.
- Audit/report files: explain data provenance, limitations, hypothesis status and inconsistencies.

The dashboard should read values from actual artifacts or APIs tied to them. Static text, a JSON file labelled Completed, or a chart visible in the UI is not proof by itself that the corresponding experiment was validly executed.

## 6. Preprocessing and Feature Engineering

### 6.1 Preprocessing order

1. Load the exact dataset file and record its source path, file hash where practical, row count and schema.
2. Identify the target column and construct X (input features) and y (labels). Remove the target and any accidental direct copy of the label from X.
3. Handle duplicates and missing values according to a documented rule; log how many rows are removed or altered.
4. Split the dataset before learning preprocessing parameters. Use stratification where appropriate for class proportions; consider group separation when records share a capture, session, domain or campaign.
5. Fit the median imputer, scaler, encoder, resampler and supervised feature selector only on the training data or within each training fold.
6. Apply the fitted transformations to validation/test data without refitting them.
7. Train the model and evaluate only on the designated held-out data. Preserve split identifiers where possible.

The recent project notes state that post-split median imputation and RobustScaler are used in the synchronized E1/E2 workflow. That guards against preprocessing statistics leaking from the test set. It does not remove leakage already embedded in precomputed context features or caused by related records sharing the split.

### 6.2 Feature families described by the research design

| Feature family | Examples from the design | Purpose |
|---|---|---|
| DNS lexical | FQDN/query length, label lengths, character distribution, Shannon entropy, digit/uppercase ratios, dictionary-word indicators. | Describe structural and character-level properties of query names. |
| DNS behavior | Query frequency/type, unique domains/name servers/IPs, response frequency, NXDOMAIN rate, unique-query ratio. | Describe repetition and request behavior over observations. |

| Feature family | Examples from the design | Purpose |
|---|---|---|
| Flow and packet | Forward/backward/total packets and bytes, packet-size mean/min/max/std, flow duration, packets/s, bytes/s. | Summarize traffic volume and packet characteristics. |
| Temporal | Inter-arrival-time mean/min/max/std, request frequency, burstiness, flow duration. | Describe timing and temporal regularity. |
| TLS/DoH | SNI, ALPN, JA3/fingerprint-type metadata, server information where observable. | Use transport metadata available without reading encrypted DNS payloads. |

### 6.3 Feature selection / explanation methods

| Method | How it works | Caution |
|---|---|---|
| Correlation analysis | Finds highly correlated features that may be redundant. | Correlation does not prove two features are semantically interchangeable. |
| Mutual information (MI) | Ranks features by statistical association with the target. | Must be fitted on training data for predictive selection; association is not causation. |
| Random Forest importance | Measures feature contribution under the model's tree-importance calculation. | Can favor features with certain distributions/cardinalities and may be unstable with correlated inputs. |
| SHAP | Attributes contributions to individual predictions; mean absolute values provide a global ranking. | Explains this model on the explained data; does not prove causality or external generalization. |

## 7. The Six Machine-Learning Models

| Model | Plain-language description | Why included | Key consideration |
|---|---|---|---|
| Logistic Regression (LR) | Fits a linear decision boundary and produces class scores/probabilities. | Fast, interpretable baseline. | Can miss complex nonlinear relationships; threshold influences F1. |
| K-Nearest Neighbors (KNN) | Classifies based on nearby training examples. | Tests whether classes form distinct local neighborhoods. | Sensitive to scaling, distance definitions, data size and repeated patterns. |
| Support Vector Machine (SVM) | Learns a separating boundary; linear/kernel configurations differ. | Classical high-dimensional classifier. | Kernel choice, scaling and sample size affect cost and behavior. |
| Random Forest (RF) | Combines many decision trees trained on varied samples/features. | Strong nonlinear tabular baseline; offers tree-based importance. | Can learn dataset/campaign-specific patterns; importance is not causal. |
| XGBoost (XGB) | Sequential gradient-boosted trees improve the objective. | Strong tabular baseline; supports gain importance and TreeSHAP. | Can fit subtle or leaked patterns; tune only on training/validation data. |
| Multi-Layer Perceptron (MLP) | A feed-forward neural network learns nonlinear feature combinations. | Neural baseline from a different model family. | Scaling, regularization, seed and architecture affect stability. |

A fair comparison requires the same data partition and the same evaluation definition within each experiment. A model comparison that mixes different samples, seeds, thresholds, feature subsets or result runs may be descriptive, but it is not a controlled head-to-head comparison.

# 8. Evaluation Metrics and How to Interpret Them

*Figure 4. The four confusion-matrix outcomes, treating exfiltration as the positive class.*

| Metric | Meaning | Why it matters |
|---|---|---|
| Precision = TP / (TP + FP) | Of records predicted as exfiltration, proportion that are positive. | Low precision means more false alarms. |
| Recall = TP / (TP + FN) | Of positive records, proportion detected. | Low recall means more missed exfiltration-labelled records. |
| F1 = 2PR / (P + R) | Harmonic mean of precision and recall. | Balances precision and recall at a chosen classification threshold. |
| FPR = FP / (FP + TN) | Fraction of benign records incorrectly flagged. | Important because security systems can overwhelm analysts with false alarms. |
| ROC-AUC | Ranking performance across classification thresholds. | High AUC does not guarantee high F1 at threshold 0.50. |
| PR-AUC | Precision–recall curve area. | Especially informative when positive examples are rare; specify the calculation convention. |
| Accuracy | Fraction of all records classified correctly. | Can be misleading when one class dominates. |
| Training/inference time | Measured model-fit and prediction cost. | Useful to compare deployment practicality; only report if actually recorded. |

## 8.1 How AUC can be high while F1 is zero

ROC-AUC assesses ranking over many thresholds; F1 uses predicted classes at one threshold. A model may assign every positive record a score greater than every negative record, producing a high ROC-AUC, while all positive scores remain below 0.50. At that threshold the model predicts every record as benign, so recall and F1 for the positive class are zero. In an actual result, label encoding, probability-column selection, score orientation, alignment and the confusion matrix must also be checked before accepting that interpretation.

# 9. E1 and E2 — Within-Dataset Detection

## 9.1 E1: UFU-Do53-EXF to UFU-Do53-EXF

Purpose: establish a conventional DNS baseline. The model is trained on a training partition from UFU-Do53-EXF and evaluated on held-out records from the same dataset. The experiment can show that features in this dataset separate its benign and exfiltration classes, but random splitting does not guarantee that test records are independent of campaign/domain patterns in training.

## 9.2 E2: UFU-DoH-EXF to UFU-DoH-EXF

Purpose: establish a baseline for DoH exfiltration using the feature representation in UFU-DoH-EXF. Since the target class is reported at about 3.1% in the published data, precision–recall behavior, recall and false-positive

rate should be examined along with F1 and ROC-AUC. A high ROC-AUC alongside a lower F1 can arise from threshold behavior or other score/prediction issues and needs metric-level diagnosis.

### 9.3 Reported synchronized 15k comparison

A synchronized-run summary shared during the project discussion reports the following F1 values for the newer 15k comparison. The values are reproduced here as reported, not independently recalculated from the local JSON files.

| Experiment / dataset | RF | KNN | XGBoost | MLP |
|---|---|---|---|---|
| E1 / UFU-Do53-EXF | 0.9983 | 0.9983 | 0.9967 | 0.9983 |
| E2 / UFU-DoH-EXF | 0.9945 | 0.9890 | 0.9946 | 0.8712 |

The latest synchronized comparison authorized KNN, XGBoost and MLP reruns alongside the existing RF baseline. LR and SVM values previously shown on the dashboard belong to earlier runs and were not necessarily evaluated on the same sample/split. Do not describe all six models as synchronized until LR and SVM have matching split artifacts.

## 10. E3 and E4 — Cross-Protocol Transfer Diagnostics

### 10.1 Intended primary experiments

| Experiment | Intended source | Intended target | Research question |
|---|---|---|---|
| E3 | UFU-Do53-EXF | UFU-DoH-EXF | Can a Do53-trained model work on DoH? |
| E4 | UFU-DoH-EXF | UFU-Do53-EXF | Can a DoH-trained model work on conventional DNS? |

The primary UFU CSV schemas are reported to share zero exact common features: Do53 has lexical/domain/context features while DoH has encrypted-flow features. Directly passing one dataset's matrix into a model trained on the other is therefore invalid unless a defensible shared representation is derived. Renaming columns or inventing values is not a scientific solution.

### 10.2 Current diagnostic proxy representation

The live dashboard runs described in the shared outputs use a seven-feature flow representation derived from CIC-Bell UDP flows and CIRA-CIC DoH/TLS flows. The shared feature names include flow duration, packet count, byte count and packet-size statistics. These runs are labelled Diagnostic only because they do not use matched flow windows extracted from the primary DNS-360 Do53 and DoH campaigns.

| E3 diagnostic model | Reported F1 | Precision | Recall | ROC-AUC | Interpretation |
|---|---|---|---|---|---|
| KNN | 0.0937 | 0.1417 | 0.0700 | 0.4637 | Poor transfer in this proxy run. |
| XGBoost | 0.6892 | 0.6089 | 0.7940 | 0.7722 | Best reported E3 F1 among the supplied runs; still proxy-only. |
| SVM | 0.1755 | 0.2230 | 0.1447 | 0.4643 | Weak transfer in this proxy run. |

| E3 diagnostic model | Reported F1 | Precision | Recall | ROC-AUC | Interpretation |
|---|---|---|---|---|---|
| MLP | 0.1749 | 0.2235 | 0.1437 | 0.3074 | Weak transfer in this proxy run. |
| LR / RF | 0.1698 / 0.0476 | see run files | see run files | varies by run | Older reported E3 outputs; verify run IDs before making a direct head-to-head comparison. |

The reported source proxy contained 15,000 flows (5,102 benign; 9,898 exfiltration), and the target proxy contained 10,000 flows (7,000 benign; 3,000 exfiltration). These are proxy flow counts, not the published UFU dataset counts. The KNN confusion matrix reported TN=5,728, FP=1,272, FN=2,790, TP=210. The XGBoost run reported TN=5,470, FP=1,530, FN=618, TP=2,382.

### 10.3 Why transfer may fail

The diagnostic notes describe an observation-unit mismatch: Do53 UDP flows may represent short query/response transactions, while DoH TCP/TLS sessions may remain open longer and multiplex multiple DNS messages. A raw five-tuple flow duration or packet-size measurement may therefore measure different behavior in the two cases. A defensible primary test needs consistent observation windows, time definitions, feature units, and ground-truth timestamp alignment.

### 10.4 E4 and high AUC with zero F1

Several E4 runs were reported with F1=0 while RF ROC-AUC varied across artifacts (one report showed about 0.9978; a later screenshot showed 0.9564). Other E4 models also showed F1=0 with different AUC values. Because run outputs differ, do not consolidate these numbers into a final model-comparison table until each value is joined to its exact run ID, predictions and configuration. Inspect the score direction, positive class, probability column, threshold, and confusion matrix.

**Defensible conclusion for E3/E4** Current E3/E4 proxy outputs are diagnostic evidence about a particular flow representation. They do not establish primary UFU-Do53 and UFU-DoH generalization. The primary transfer claim remains unresolved until compatible, consistently derived primary features are available.

## 11. E5 and E6 — Feature Studies

### 11.1 E5: Feature-group ablation

Ablation removes one feature group at a time and compares performance with a baseline. The research design proposes comparisons such as lexical-only, protocol fields, domain context, heuristics, flow-only, temporal/common features where the dataset supports them. The goal is to estimate which feature families are useful to the fitted model.

If F1 does not change after removing a group, possible interpretations include redundancy among groups, one remaining feature group being sufficient, an insensitive test setup, or an implementation problem that failed to remove the requested columns. Verify the actual selected column list for every run. A performance drop should never be written into the report unless the measured result shows it.

### 11.2 E6: Top-K feature selection

E6 compares a full feature set with compact subsets such as the top 10, 15 or 20 features. The intended trade-off is performance versus fewer inputs, reduced complexity, potentially lower inference cost and easier interpretation. Feature rankings must be calculated using training data only. Each subset must actually train a new model and use the same untouched test split.

**Status conflict to resolve** The shared history contains different E5/E6 states: some dashboard views reported RF F1=1.0000, an earlier artifact summary said all ablations had zero F1 drop, and a later experiment screen marked E5/E6 “Not run.” This document therefore treats E5/E6 as requiring artifact-to-dashboard reconciliation before their results are considered verified.

## 12. E7 and E8 — External Evaluation

### 12.1 E7: External benchmarks

E7 uses two external datasets, evaluated separately because they use different traffic types and feature representations.

| Benchmark | Reported setup | Reported score summary | Important caveat |
|---|---|---|---|
| CIRA-CIC-DoHBrw-2020 L2 | l2-benign.csv vs l2-malicious.csv; 995 deduplicated flows; 796 train / 199 test in one run. | Dashboard report: RF F1=0.9950, ROC-AUC=0.9998. | Represents benign vs malicious DoH/tunneling labels as defined in the dataset; not automatically a pure exfiltration task. |
| CIC-Bell-DNS-EXF-2021 Stateless | 12 numeric lexical features; one reported sample of 1,000 deduplicated records, 800 train / 200 test. | Dashboard report: F1=0.8412, ROC-AUC=0.8220. | Verify exact file list, folder labels, split, positive-class meaning and run-specific artifact. |
| CIC-Bell-DNS-EXF-2021 Stateful | 20 numeric RR/TTL-related features; one report says 1,000 deduplicated records. | No single confirmed score stated in this guide. | Exact split and result artifact need confirmation. |

External benchmark evaluation asks how a model performs on an additional dataset under that dataset’s own feature and label definitions. It is not the same as zero-shot transfer from a primary UFU model. If RF was trained and tested independently on CIRA-CIC, that demonstrates performance on its own split—not that a UFU-trained model transfers to CIRA-CIC.

### 12.2 E8: DNSTunnel2026

E8 is intended to use DNSTunnel2026 as a further external evaluation. The dashboard summaries report an external dataset with 38 features and train/test files, and show within-dataset results with limitations. One reported dashboard snapshot lists LR F1=0.9974, RF F1=0.9949, and XGBoost F1=0.9971. Treat these as reported run outputs pending artifact verification.

The transfer part of E8 was reported as blocked because the primary Do53 feature schema and DNSTunnel2026 window-level schema did not align. A valid evaluation of DNSTunnel2026’s own train/test split can still exist even while transfer from a UFU-trained model is blocked. These should have separate experiment names, metrics and status labels.

# 13. Explainability: Feature Importance, SHAP, and Plots

## 13.1 What SHAP means

SHAP (SHapley Additive exPlanations) attributes parts of a model output to individual input features. A local explanation describes one prediction; a global summary aggregates contributions across a set of records. Mean absolute SHAP values show the average magnitude of a feature's contribution, not whether high values push a prediction toward benign or exfiltration.

## 13.2 What the supplied E1/E2 SHAP PDF shows

| Dataset | Prominent features visible in supplied plots | Interpretation |
|---|---|---|
| E1: UFU-Do53-EXF | var_ttl, num_queries_per_domain, contains_dictionary_words, num_unique_ttl, query_length, mean_ttl. | The fitted model places substantial importance on TTL variation and domain/query context, with lexical indicators also contributing. |
| E2: UFU-DoH-EXF | src2dst_mean_time, bidirectional_mean_time, src2dst_min_length, bidirectional_min_time, bidirectional_max_time, src2dst_mean_length, bidirectional_stddev_time. | The fitted model relies heavily on directional/bidirectional timing and packet-length characteristics that are available in the DoH feature representation. |

The submitted two-page visual summary displayed a held-out SHAP subset of n=200. Approximate values read from that image included Do53 mean absolute SHAP of about 5.77 for var_ttl, 1.68 for num_queries_per_domain and 1.42 for contains_dictionary_words. For DoH, the approximate leading magnitudes were 3.51 for src2dst_mean_time, 1.90 for bidirectional_mean_time and 1.26 for src2dst_min_length. These are visual readings, not independently verified raw SHAP values; use saved attribution data for final paper numbers.

## 13.3 Why native feature gain and SHAP can differ

XGBoost native gain summarizes improvement associated with tree splits during training. SHAP summarizes feature contributions to predictions on the explained data. They are different measures and may use different data partitions. Correlated features can share or exchange importance, so rank differences are not automatically a contradiction.

## 13.4 Plot guide

| Plot | What it shows | Common mistake to avoid |
|---|---|---|
| Class distribution | Counts/percentages of benign and positive records. | Ignoring class imbalance or using published totals for a sampled run. |
| Confusion matrix | TP, FP, TN, FN at one threshold. | Calling a model effective from accuracy alone. |
| ROC curve | TPR vs FPR across score thresholds. | Treating AUC as a guarantee of good default-threshold predictions. |
| Precision–recall curve | Precision vs recall across thresholds. | Comparing curves without noting positive-class prevalence and split. |
| Feature importance | A model-specific ranking of influential predictors. | Interpreting importance as causation or cross-dataset transfer. |

| Plot | What it shows | Common mistake to avoid |
|---|---|---|
| SHAP summary/bar plot | Contribution magnitudes/directions (depending on plot type) across explained records. | Assuming absolute SHAP bar charts reveal direction or causality. |
| Ablation plot | Metric changes when groups are excluded. | Trusting it unless the intended columns were truly removed. |
| Top-K plot | Performance as feature count is reduced. | Selecting features using test labels or reusing baseline scores. |

# 14. Current Reported Results and Status Summary

The table below consolidates the project status as described in shared dashboard and run summaries. It is a handover snapshot, not a fresh execution of the local repository. Where artifacts or runs conflict, the status is qualified and the underlying files should be checked before final submission.

| Experiment | Reported execution/status | What can be said safely |
|---|---|---|
| E1 | Within-dataset Do53; synchronized comparison reported for RF/KNN/XGB/MLP. | Random-split F1 values are very high. Older LR/SVM runs are not necessarily on the same split. Generalization requires group-separated evidence. |
| E2 | Within-dataset DoH; synchronized comparison reported for RF/KNN/XGB/MLP. | F1 remains high for RF/KNN/XGB and lower for MLP in the synchronized summary. The positive class is rare in published data; PR-AUC/recall/FPR matter. |
| E3 | Several live runs executed as Diagnostic only using CIC-Bell UDP to CIRA-CIC DoH proxy flows. | Metrics describe that proxy flow setup, not primary UFU cross-protocol transfer. |
| E4 | Diagnostic proxy runs reported; F1=0 in multiple results, while AUC varies by run. | Inspect exact run IDs, labels, probability direction, test predictions and threshold. Do not merge conflicting AUC values. |
| E5 | Status/metrics conflict across dashboard and artifacts. | Treat as unverified until current run artifacts show feature groups, baseline, actual fits, same split and metric outputs. |
| E6 | Status/metrics conflict across dashboard and artifacts. | Treat as unverified until train-only feature ranking and fresh models for each Top-K subset are confirmed. |
| E7 | External CIRA/CIC-Bell evaluations reported with separate metrics. | Verify input files, task labels, sampling, test split and result files. Do not claim UFU transfer from independent benchmark runs. |
| E8 | DNSTunnel2026 within-dataset scores reported; transfer audit blocked. | Report within-dataset and transfer results separately; verify the 38-feature schema and labels. |

## 14.1 Synchronized E1/E2 F1 values shared during the project

| Model | E1 Do53 F1 | E2 DoH F1 | Comparison note |
|---|---|---|---|
| Random Forest | 0.9983 | 0.9945 | Existing 15k reference baseline in the synchronized summary. |
| KNN | 0.9983 | 0.9890 | New synchronized 15k runs. |

| Model | E1 Do53 F1 | E2 DoH F1 | Comparison note |
|---|---|---|---|
| XGBoost | 0.9967 | 0.9946 | New synchronized 15k runs. |
| MLP | 0.9983 | 0.8712 | New synchronized 15k runs. |
| Logistic Regression | 0.9985 (older run) | 0.6820 (older run) | Not rerun in the synchronized task; do not label split-matched. |
| SVM | 0.9990 (older run) | 0.7263 (older run) | Not rerun in the synchronized task; do not label split-matched. |

The synchronized F1 values above are from the run summary shared in the conversation. Before using them in a submitted table, cross-check each with its run configuration, test indices, predictions CSV, and metrics JSON. When a score is rounded to four decimals, keep the underlying stored value for calculations.

### 14.2 Why some results conflict

- Archived 12k and synchronized 15k runs can produce different scores because the sample and split differ.
- The dashboard, report generator, metrics JSONs, and experiment details have previously shown inconsistent statuses or numbers.
- E3 has multiple model-specific proxy run IDs and different outputs; they may not share the same split or configuration.
- E4 has more than one reported ROC-AUC value for the same model across different run summaries.
- E5/E6 have been shown as both completed with perfect RF results and not run in different views.
- E7 has a combined/external card and separate benchmark results; separate dataset values should be preserved.
- E8 within-dataset evaluation and transfer validation are distinct and should not share a single status or score.

**Rule for Table V** Use one clearly identified run per cell or label the table as a mixed-run status summary. A dash means “not available from the evidence used,” not zero. E3/E4 must be marked as diagnostic proxy results until the primary transfer setup is validated.

## 15. Threats to Validity and Scientific Integrity

### 15.1 Data leakage

Data leakage occurs when information unavailable at prediction time influences the trained model or evaluation. It can happen through direct target leakage, identifiers or metadata that reveal labels, fitting preprocessing on the whole dataset, feature selection using test labels, duplicate records across the split, or correlated examples from the same source group. Correctly fitting a scaler on train-only data addresses only one subset of these risks.

### 15.2 Group-separated evaluation

Random-row splits can put examples from the same domain, campaign, session, capture or generator in both training and testing. A group-separated split holds out complete groups, which better tests whether the model transfers to new sources of variation. The grouping key must match the scientific claim: unseen domains, sessions, campaigns and future time periods are related but not identical generalization tests.

A shared audit summary reports that Do53 group-separated F1 values ranged approximately from 0.0041 to 0.4657 across reported configurations, while DoH session-group F1 was around 0.6365 for several models. These ranges illustrate a substantial generalization gap but must be tied to exact artifact names/configurations before publication.

### 15.3 Data provenance and synthetic data

Synthetic data can test whether loaders, APIs, charts and experiment runners function. It cannot establish that a detector works on real network traffic. Each run should identify whether its data is authentic, derived from packet captures, synthetic, or a proxy constructed from another benchmark. Synthetic rows or labels must never be presented as genuine external benchmark samples.

### 15.4 Small benchmark samples

The reported CIRA L2 test set has 199 flows in one evaluation, and the reported CIC-Bell stateless test set has 200. Metrics from small test sets can vary substantially with a small number of errors. The report should include class counts, confusion matrices, and uncertainty analysis where feasible. Deduplication should be described, and related records should not cross the split unnoticed.

### 15.5 Evidence hierarchy

| Label | Meaning |
|---|---|
| Verified | Input, labels, run configuration, predictions and metrics have been checked for the stated claim. |
| Executed, needs validation | The code ran and produced outputs, but one or more validity checks remain unresolved. |
| Diagnostic/proxy only | An exploratory setup ran, but it does not establish the intended primary claim. |
| Blocked | Required data, labels, or feature compatibility are unavailable for the intended experiment. |
| Not run | There is no adequate evidence of execution. |
| Synthetic demonstration | Pipeline run uses generated data and is not real-traffic validation. |

## 16. Running, Reviewing, and Reporting the Project

### 16.1 What teammates should inspect for any run

8. Find the exact run ID and experiment ID.
9. Open the run configuration JSON and record the dataset path, model, sample limit, seed, feature list and split strategy.
10. Confirm the input provenance and label meanings.
11. Check the prediction CSV has aligned true labels, predicted classes and prediction scores where applicable.
12. Recalculate metrics from predictions if possible; do not rely only on dashboard text.
13. Open the confusion matrix, ROC/PR curves and relevant explanation output.
14. Compare the displayed dashboard values with the metric JSON and report output.

15. Record the status as verified, executed/unverified, diagnostic, blocked, not run, or synthetic demonstration.

### 16.2 Known local workflow and commands

The development workflow shared previously used a Windows PowerShell environment, the `icsproject` directory, a Python virtual environment, a backend API, and a Vite frontend served at localhost port 5173. Confirm the current README and package scripts before relying on the exact command sequence.

| Task | Command / guidance |
|---|---|
| Open project folder | `cd "C:\Users\Sharayu Pathare\Desktop\icsproject"` |
| Check Python | `python --version` and `python -c "import sys; print(sys.executable)"` |
| Backend (previously used) | `python -m uvicorn backend.server:app --host 127.0.0.1 --port 8000 --reload` — verify `backend.server:app` still matches the current code. |
| Frontend | Run the dev script defined in `package.json` (often `npm run dev` for Vite); verify the current scripts rather than guessing. |
| Research report (previously used) | `python -c "import backend.report_generator as rg; print(rg.generate_research_report())"` |
| Experiment execution | Use the current dashboard live-run workflow or a confirmed CLI entry point. The earlier `python -m backend.experiments run ...` command returned silently, so do not assume it executes experiments unless the current module documents that CLI. |

### 16.3 Checkpoint and change safety

The project history reports an on-disk snapshot at `checkpoints/checkpoint_20261013_1008/`, an integrity manifest `CHECKPOINT_MANIFEST.json`, and a restore guide `checkpoint.md`, with 235 files reported as SHA-256 verified. This is a useful safeguard, but teams should check the manifest coverage and restore instructions before changing experiment logic. New audit runs should use separate run-specific output paths; historical metrics should not be silently overwritten.

## 17. Viva Preparation and Glossary

### 17.1 Short project explanation (30 seconds)

**Suggested explanation** Our project compares six machine-learning models for detecting DNS exfiltration-labelled traffic in conventional DNS and DNS-over-HTTPS datasets. We evaluate within-dataset performance, investigate cross-protocol and external generalization, test feature groups and compact feature sets, and use SHAP to understand model decisions. A key finding from our audit is that near-perfect random-split scores can fall under group-separated evaluation, so generalization and data provenance are critical.

## 17.2 Likely viva questions

| Question | Answer |
|---|---|
| Why use DNS for exfiltration? | DNS is widely needed, so it can be attractive to abuse as a covert path for encoded data. |
| Why is DoH harder to inspect? | DNS messages are carried inside HTTPS, hiding the query payload from a passive observer without decryption. |
| Why compare six classifiers? | They represent different model families and provide a broader performance/complexity comparison. |
| Why report PR-AUC as well as ROC-AUC? | PR-AUC is often more informative when the positive class is rare. |
| Can F1 be 1.0? | Yes, on a particular test set. It does not prove universal generalization or rule out leakage. |
| Why can ROC-AUC be high but F1=0? | Good score ranking can coexist with a threshold that predicts no positive examples; also check label and score alignment. |
| Why is E3 diagnostic only? | The current proxy features originate from different external traffic sources, not matched flow windows from both primary UFU datasets. |
| What does E5 do? | Removes feature groups to measure how model performance changes. |
| What does E6 do? | Compares full and top-K feature sets to study compactness/performance trade-offs. |
| What does SHAP tell us? | How the fitted model attributes contributions to predictions on the explained samples; it does not prove causality. |
| Why are datasets imbalanced? | Benign records substantially outnumber exfiltration records in both primary datasets. |
| What is the biggest limitation? | Generalization and feature compatibility; random-split results can exploit dataset-specific patterns. |
| Does a CIRA result prove exfiltration detection? | Only if the benchmark label/task corresponds to the exfiltration claim; malicious DoH and exfiltration are not identical. |
| Why can E8 be partly completed? | Within-dataset train/test evaluation may be possible while primary-to-DNSTunnel transfer is blocked by feature incompatibility. |
| What makes an experiment reproducible? | Traceable inputs, labels, split indices, seed, configuration, predictions, metrics and artifact provenance. |

## 17.3 Glossary

| Term | Meaning |
|---|---|
| Ablation | Experiment that removes a feature group to measure its contribution. |
| AUC | Area under a curve; ROC-AUC measures ranking over thresholds. |
| Class imbalance | One class appears much more often than another. |
| Cross-dataset evaluation | Testing on a dataset different from the training dataset. |
| Domain/campaign group | Records related to a shared domain, attack campaign, or collection source. |

| Term | Meaning |
|---|---|
| Feature leakage | Input contains information that makes the label easy to infer in a way unavailable for intended deployment. |
| PR-AUC | Area under precision–recall curve. |
| RobustScaler | Transforms features using statistics based on median and quantile range. |
| SHAP | Framework for attributing model output contributions to input features. |
| Stratified split | Split that attempts to preserve class proportions. |
| Zero-shot transfer | Train on a source dataset and test on a target without target-data retraining. |

# 18. References

The following entries reproduce the reference list included in the project research-paper draft. Check each publisher record for final citation metadata and access information before submitting a formal paper.

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## Final Guidance for the Team

When presenting the project, always separate (1) the research design, (2) the code that exists, (3) the experiments that actually ran, and (4) the claims supported by valid evidence. The same dashboard may contain archived random-split metrics, synchronized runs, diagnostic proxies, external benchmark results, and blocked transfer evaluations. A good explanation identifies each by its run ID and data source.

The most defensible overall message is: the project compares six classifiers across conventional DNS and DoH exfiltration datasets, explores feature relevance using ablation and SHAP, and investigates whether good within-dataset results generalize. The audit suggests the generalization question is central because performance can drop substantially under group-separated testing, while current primary cross-protocol transfer remains constrained by incompatible features and incomplete PCAP archives.

**Before formal submission** Reconcile the E3/E4 run-specific AUC discrepancies, E5/E6 status conflict, benchmark sample counts/splits, DoH PCAP missing-part count, and every final Table V value against the current metric JSON, prediction CSV and run configuration. Those are evidence-management issues; do not resolve them by choosing the best-looking values.